Heterogeneous Pricing of Supply Chain Risk in Production Networks

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Abstract

This paper examines whether supply chain risk transmitted through production networks is priced in equity markets. We construct a novel firm-level measure of Network Supply Chain Risk (NSCR) by combining input-output linkages from the Bureau of Economic Analysis with macroeconomic shocks from the Federal Reserve Economic Data database. Using a panel of U.S. firms from 2018 to 2024, we test whether exposure to upstream input shocks affects stock returns. While baseline effects weaken after controlling for standard asset pricing factors, we find strong evidence of heterogeneity. Supply chain risk negatively affects low-exposure firms but positively affects high-exposure firms, consistent with risk compensation and adaptation. These findings contribute to the growing literature on production networks and asset pricing by demonstrating that supply chain risk is not uniformly priced but depends critically on firm exposure, industry structure, and shock magnitude.

JEL Classification: G12; L14; E44; F14

Keywords: production networks; supply chain risk; asset pricing; firm heterogeneity; equity returns; production linkages

1. Introduction

Global production has become increasingly interconnected, with firms embedded in complex supply chain networks that transmit shocks across industries and geographic boundaries. Events such as the COVID-19 pandemic, geopolitical tensions, and commodity price volatility have highlighted the vulnerability of these networks and their potential impact on firm performance (Carvalho et al., 2021; Baldwin, 2016). Despite this, the role of supply chain risk in asset pricing remains insufficiently understood. The COVID-19 pandemic further highlighted the role of global supply chains in transmitting shocks across countries and sectors (Bonadio et al., 2021).

Traditional asset pricing models largely focus on aggregate risk factors, such as market, size, and value (Fama and French, 1993), while abstracting from the network structure of production. However, recent advances in the study of production networks suggest that firm-level outcomes are shaped not only by idiosyncratic shocks but also by upstream and downstream linkages (Acemoglu et al., 2012; Carvalho, 2014). These networks can amplify and propagate shocks, implying that exposure to input markets—such as oil, metals, and chemicals—may represent an economically meaningful source of risk.

This paper introduces a novel framework to quantify and test the pricing of such risks. We construct a firm-level measure of Network Supply Chain Risk (NSCR) by combining industry-level input-output weights from Bureau of Economic Analysis tables with macroeconomic shocks derived from Federal Reserve Economic Data time series. The resulting measure captures the extent to which firms are exposed to fluctuations in key upstream inputs through their production networks.

Our empirical analysis yields three main findings. First, baseline effects weaken after controlling for standard asset pricing factors, indicating that supply chain risk is not an independent systematic risk factor. Second, we document substantial heterogeneity in this relationship. When firms are partitioned based on their level of exposure, we find that the negative effect of NSCR is concentrated among low-exposure firms, while high-exposure firms exhibit a positive sensitivity. This pattern is consistent with models in which highly exposed firms either command a risk premium or possess greater adaptability to supply chain shocks. Third, our results show that supply chain risk is not uniformly priced but exhibits strong heterogeneity across firms. Additionally, we find that the pricing of NSCR varies across industries, with the strongest effects observed in energy-related sectors.

1.1. Contribution

This paper contributes to three strands of literature. By linking input-output structures to financial outcomes, we extend the literature showing that network linkages amplify shocks (Acemoglu et al., 2012; Carvalho, 2014). We demonstrate that supply chain risk is heterogeneously priced across firms, showing that its impact depends critically on firm exposure rather than constituting a standalone risk factor. While prior studies examine operational impacts of supply disruptions, we provide direct evidence on their pricing in equity markets.

2. Literature Review

This paper sits at the intersection of three major strands of literature: (i) production networks and shock propagation, (ii) asset pricing and risk factors, and (iii) supply chain risk and firm performance. We discuss each in turn and highlight how our contribution differs.

2.1. Production Networks and Shock Transmission

A growing body of research emphasizes that economic activity is organized through complex production networks, where firms and industries are connected through input-output relationships. In this framework, shocks affecting one sector can propagate and amplify across the entire economy. Seminal work by Acemoglu et al. (2012) shows that micro-level shocks can generate aggregate fluctuations when the network structure is sufficiently asymmetric. Similarly, Carvalho (2014) demonstrates that production networks play a central role in transmitting sectoral shocks, particularly when certain industries serve as key input suppliers.

Subsequent studies extend this framework to firm-level data. For example, Baqaee and Farhi (2020) show that nonlinearities in production networks can amplify shocks beyond what standard linear models predict. These findings imply that exposure to upstream industries—such as energy, metals, and chemicals—can significantly influence firm-level outcomes (Barrot and Sauvagnat, 2016; Carvalho et al., 2021). Empirical evidence also highlights the role of input linkages in propagating shocks across firms and industries (Boehm et al., 2019; Atalay et al., 2011). Despite these advances, the literature has primarily focused on macroeconomic implications, such as output volatility and aggregate productivity, leaving relatively less attention to how production network exposure affects financial markets.

2.2. Asset Pricing and Risk Factors

Traditional asset pricing models, beginning with Fama and French (1993), explain cross-sectional variation in returns using factors such as market risk, size, and value. Subsequent extensions incorporate momentum, profitability, and investment factors, among others (Carhart, 1997; Fama and French, 2015; Jegadeesh and Titman, 1993). However, these models largely abstract from the real-side structure of the economy, treating firms as isolated units rather than nodes in a network.

More recent research has begun to incorporate real economic linkages into asset pricing frameworks. Studies show that exposure to macroeconomic variables—such as commodity prices and aggregate volatility—can influence expected returns (Chen et al., 1986). Commodity price fluctuations have also been shown to affect stock returns, particularly for firms with exposure to input markets (Ready, 2018). Another related strand examines how firm characteristics tied to production—such as operating leverage or input costs—affect risk and returns (Novy-Marx, 2013; Gourio, 2012). Yet, these approaches typically do not account for inter-industry dependencies captured by input-output networks. Recent research incorporating network structure into asset pricing includes Ahern (2013), who shows that firms' position within economic networks affects expected returns, and Herskovic (2018), who demonstrates that production network linkages can generate systematic risk.

2.3. Supply Chain Risk and Firm Outcomes

The third relevant strand focuses on supply chain disruptions and their impact on firm performance. Events such as natural disasters, trade conflicts, and pandemics have highlighted the vulnerability of global supply chains (Hendricks and Singhal, 2005; Tang, 2006). Empirical studies show that disruptions to suppliers can negatively affect firm performance, profitability, and stock prices (Barrot and Sauvagnat, 2016; Hendricks and Singhal, 2005). Similarly, research on global value chains demonstrates that shocks in upstream industries can propagate across countries and sectors (Baldwin, 2016; Antràs and Chor, 2013). More recently, the COVID-19 pandemic has spurred renewed interest in supply chain resilience (Carvalho et al., 2021; Tang, 2006; Zsidisin and Ritchie, 2009). However, most of this literature focuses on event-based analysis or operational outcomes, rather than systematic pricing in financial markets.

2.4. Gap in the Literature

Taken together, existing research highlights the importance of production networks in transmitting shocks, the role of risk factors in explaining asset returns, and the economic consequences of supply chain disruptions. Yet, a key gap remains: there is no unified framework that links production network exposure to asset pricing through a measurable, firm-level risk factor. This paper fills this gap by constructing a novel risk measure, bridging the real and financial sectors, documenting heterogeneous effects across firms, and providing new empirical evidence using a panel of U.S. firms.

3. Data and Construction of Network Supply Chain Risk

This section describes the data sources and the construction of the key variable used in the analysis: Network Supply Chain Risk (NSCR). The dataset combines firm-level stock returns, industry-level input-output relationships, and macroeconomic shock variables.

3.1. Stock Return Data

Firm-level financial data are collected for a sample of publicly listed companies over the period January 2018 to December 2024. Monthly stock returns are computed using closing price data. For each firm i and month t, returns are defined as:

$$R_{it} = (P_{it} - P_{it-1}) / P_{it-1}$$

where P_{it} is the closing price of firm i at time t. The final dataset consists of approximately 50 firms across multiple industries, including technology, automobile, energy, retail, industrials, and healthcare. Each firm is mapped to a primary industry classification, which is later used to link firm-level data with industry-level input-output weights.

3.2. Input-Output Data

To capture supply chain linkages, we use industry-level input-output tables from the Bureau of Economic Analysis (BEA). These tables provide information on how industries depend on upstream inputs, such as oil, metals, energy, and chemicals. From the BEA data, we construct a normalized weight matrix w_{ij}, where each element represents the dependence of industry i on input j. For each industry, the weights are normalized such that $\Sigma_j w_{ij} = 1$. These weights reflect the relative importance of different input categories in the production process of each industry.

3.3. Macroeconomic Shock Variables

Macroeconomic shocks are constructed using time series data from the Federal Reserve Economic Data (FRED) database. We focus on variables that proxy for fluctuations in key input markets: oil prices, metal prices, energy indices, and chemical price indices. Each series is transformed into monthly percentage changes to capture shocks:

$$Shock_{jt} = (X_{jt} - X_{jt-1}) / X_{jt-1}$$

where X_{jt} represents the value of macro variable j at time t. These transformations ensure that the variables reflect unexpected changes rather than levels.

3.4. Construction of Network Supply Chain Risk

We combine the input-output weights with macroeconomic shocks to construct the key variable, Network Supply Chain Risk (NSCR):

$$NSCR_{it} = \Sigma_j w_{ij} \times Shock_{jt}$$

where w_{ij} is the weight of input j for industry i and $Shock_{jt}$ is the macroeconomic shock to input j at time t. This measure captures the extent to which a firm is exposed to fluctuations in upstream input markets through its industry's production network. A higher $NSCR_{it}$ indicates greater exposure to adverse input shocks. The measure is time-varying and firm-specific via industry mapping, reflecting both network structure (weights) and macroeconomic volatility (shocks).

3.5. Lag Structure

To mitigate concerns of reverse causality, we use a lagged version of NSCR in the empirical analysis: $NSCR_{i,t-1}$. This ensures that supply chain risk is measured prior to the realization of returns.

3.6. Final Dataset

The final panel dataset is described in Table A1 in the appendix. The final panel dataset contains firm-level monthly observations suitable for regression analysis.

Table A1 Variable Descriptions

Variable	**Description**
Firm	Firm identifier
Date	Monthly time index
Return	Monthly stock return
NSCR	Network Supply Chain Risk
Industry	Industry classification

Notes: This table describes the variables in the final panel dataset.

4. Empirical Methodology

This section outlines the econometric framework used to test whether network-based supply chain risk is priced in equity returns. We begin with a baseline specification and then extend the model to capture heterogeneity and contextual effects.

4.1. Baseline Specification

To examine whether supply chain risk affects stock returns, we estimate the following panel regression:

$$R_{it} = \alpha + \beta\, NSCR_{i,t-1} + \mu_i + \varepsilon_{it}$$

where R_{it} is the return of firm i in month t, $NSCR_{i,t-1}$ is the lagged network supply chain risk, μ_i represents firm fixed effects, and ε_{it} is the error term. The coefficient β captures the sensitivity of firm returns to supply chain risk. A negative value implies that firms exposed to adverse upstream shocks experience lower returns, while a positive value suggests compensation for bearing such risk.

4.2. Fixed Effects Specification

We include firm fixed effects (μ_i) to control for time-invariant firm characteristics such as size, business model, capital structure, and industry positioning. This ensures that identification comes from within-firm variation over time, rather than cross-sectional differences. Industry effects are already embedded within firm fixed effects, as each firm belongs to a specific industry, so including both would introduce perfect multicollinearity.

4.3. Lag Structure and Identification

To mitigate reverse causality concerns, we use a lagged measure of NSCR: $NSCR_{i,t-1}$. Identification relies on two sources of exogenous variation. First, input price changes (e.g., oil, metals) are largely driven by global factors and are plausibly exogenous to individual firms. Second, input-output weights from BEA reflect structural production relationships that are slow-moving and not influenced by short-term financial outcomes. Together, these components generate variation in NSCR that is plausibly exogenous to firm-level returns.

4.4. Heterogeneity Analysis

To examine whether the effect of supply chain risk varies across firms, we estimate an interaction model:

$$R_{it} = \alpha + \beta_1\, NSCR_{i,t-1} + \beta_2(NSCR_{i,t-1} \times High_i) + \mu_i + \varepsilon_{it}$$

where $High_i$ is an indicator equal to 1 for firms with above-median NSCR exposure. The coefficient β_1 captures the effect for low-exposure firms, while $\beta_1 + \beta_2$ captures the effect for high-exposure firms. This specification allows us to test whether supply chain risk is priced differently depending on exposure intensity.

4.5. Fama–French Factor Controls

We augment our baseline specification by including standard Fama–French (1993) factors, namely the market factor (MKT), size (SMB), and value (HML), to ensure that the observed effects are not driven by known sources of systematic risk. After controlling for Fama–French factors, the baseline effect of NSCR becomes statistically insignificant, indicating that supply chain risk is not an independent risk factor. This motivates our focus on heterogeneity as the primary contribution of this paper.

4.6. Crisis Interaction Model

To test whether supply chain risk behaves differently during periods of economic disruption, we estimate:

$$R_{it} = \alpha + \beta_1\, NSCR_{i,t-1} + \beta_2(NSCR_{i,t-1} \times Crisis_t) + \mu_i + \varepsilon_{it}$$

where $Crisis_t$ is an indicator for the COVID-19 period (2020–2021). The coefficient β_1 captures the effect during normal periods, while β_2 captures the additional effect during the crisis.

4.7. Industry Interaction Model

To examine sectoral differences, we estimate a model that allows the effect of NSCR to vary across industries, capturing differences in input dependence, exposure to commodity markets, and supply chain structure:

$$R_{it} = \alpha + \beta\ NSCR_{i,t-1} + \Sigma_k\ \gamma_k(NSCR_{i,t-1} \times Industry_k) + \varepsilon_{it}$$

4.8. Estimation Method

All regressions are estimated using ordinary least squares (OLS) with firm-clustered standard errors (Petersen, 2009). Clustering at the firm level accounts for serial correlation in firm returns and heteroskedasticity across firms.

4.9. Portfolio Analysis

In addition to regression-based evidence, we construct portfolios based on NSCR exposure. Firms are sorted into high and low NSCR groups, monthly portfolio returns are computed, and a long–short portfolio is formed:

$$R_t^{LL} = R_t^{High} - R_t^{Low}$$

A positive long–short return suggests that high NSCR firms outperform low NSCR firms, providing an alternative test of whether supply chain risk is priced.

5. Empirical Results

This section presents the main findings on the relationship between network supply chain risk (NSCR) and stock returns. We begin with the baseline specification and then examine heterogeneity, crisis dynamics, and industry-specific patterns.

5.1. Baseline Results

Table 1 reports the baseline regression results estimating the effect of lagged network supply chain risk on firm-level returns. The coefficient on $NSCR_{i,t-1}$ is negative and statistically significant at conventional levels. Specifically, the estimated coefficient is approximately −0.059, indicating that higher exposure to supply chain shocks is associated with lower subsequent stock returns. This finding suggests that firms with greater exposure to upstream input shocks—such as fluctuations in oil, metals, and chemical prices—tend to underperform in equity markets, consistent with supply chain disruptions increasing production costs, reducing expected cash flows, and introducing operational uncertainty (Hendricks and Singhal, 2005; Barrot and Sauvagnat, 2016).

Table 1 Baseline Results: Effect of NSCR on Stock Returns

Variables	**(1)**
NSCR_lag	−0.0587***
	(0.021)
Firm Fixed Effects	Yes
Observations	3,690
R^2	0.042

Notes: Dependent variable is the monthly firm-level stock return. Standard errors (in parentheses) are clustered at the firm level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

5.2. Heterogeneity in Risk Pricing

To examine whether the effect of NSCR varies across firms, we estimate an interaction model distinguishing between high- and low-exposure firms. Table 2 presents the results. The coefficient on $NSCR_{i,t-1}$ is −0.0873 and statistically significant, while the interaction term $NSCR_{i,t-1} \times High_i$ is positive ($\beta = +0.1178$, $p < 0.05$). This pattern is consistent with risk compensation and adaptation among high-exposure firms.

Table 2 Heterogeneity Analysis: High vs. Low NSCR Exposure

Variables	**(1)**
NSCR_lag	−0.0873**
	(0.028)
NSCR_lag × High_NS	0.1178**
	(0.049)
Firm Fixed Effects	Yes
Observations	3,690
R^2	0.057

Notes: High_NS is an indicator equal to 1 for firms with above-median NSCR exposure. Standard errors clustered at the firm level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

These results imply that low-exposure firms experience a strong negative impact from supply chain risk, while high-exposure firms exhibit a positive sensitivity. One possible explanation is that highly exposed firms are better adapted to input volatility, can pass costs to consumers, or command a risk premium for bearing supply chain risk. This pattern is consistent with models of heterogeneous risk exposure and information frictions (Merton, 1987).

5.2.1. Economic Magnitude

To assess the economic significance of the heterogeneity results, we compute the implied return effects of a one standard deviation increase in NSCR ($\sigma = 0.0618$). For low-exposure firms, this translates to a reduction in returns of approximately 0.54% per month, equivalent to approximately 6.5% annually. For high-exposure firms, the net effect is positive, amounting to an increase of approximately 0.19% per month, or approximately 2.3% annually. These magnitudes are economically meaningful and underscore the practical relevance of our heterogeneity findings for portfolio construction and risk management.

5.3. Fama–French Controlled Results

Table 3 presents results after controlling for Fama–French (1993) factors. The baseline effect of NSCR becomes statistically insignificant ($p = 0.341$), indicating that supply chain risk is not an independent risk factor. By contrast, the market factor (MKT) retains strong statistical significance, confirming the robustness of standard asset pricing factors. These results motivate our focus on heterogeneous pricing as the primary contribution of this paper.

Table 3 Fama–French Controlled Results

Variable	**Coefficient**
NSCR_lag	−0.0312 ($p = 0.341$)
MKT	0.8743*** ($p < 0.001$)
SMB	0.1254 ($p = 0.218$)
HML	−0.0871 ($p = 0.412$)

Notes: After including Fama–French (1993) factors, NSCR's baseline effect becomes insignificant ($p = 0.341$), while MKT is strongly significant ($p < 0.001$). Firm fixed effects included. Standard errors clustered at the firm level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

5.4. Crisis Period Analysis

Table 4 investigates whether the pricing of NSCR differs during periods of economic disruption, focusing on the COVID-19 pandemic. The interaction term with the crisis indicator is not statistically significant, suggesting that the effect of supply chain risk on returns is persistent and not significantly amplified during crisis periods. In other words, supply chain risk appears to be a structural factor, rather than one that only becomes relevant during extreme events.

Table 4 Crisis Period Analysis: NSCR Effects During COVID-19

Variables	**(1)**
NSCR_lag	−0.1102**

	(0.043)
NSCR_lag × Crisis	0.018
	(0.060)
Firm Fixed Effects	Yes
Observations	3,690
R^2	0.044

Notes: Crisis is an indicator for the COVID-19 period (2020–2021). Standard errors clustered at the firm level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

5.5. Industry-Level Differences

Table 5 examines whether the effect of NSCR varies across industries. The interaction between NSCR and the energy sector is negative and statistically significant, while other industries do not exhibit strong differential effects. This implies that supply chain risk is most strongly priced in industries directly exposed to input markets, particularly energy-related sectors, consistent with the intuition that industries heavily reliant on volatile inputs are more sensitive to supply chain disruptions.

Table 5 Industry Effects: Energy Sector Interaction

Variables	**(1)**
NSCR_lag	−0.040
	(0.030)
NSCR_lag × Energy	−0.5434**
	(0.210)
Firm Fixed Effects	No
Observations	3,690
R^2	0.038

Notes: Energy is an indicator for energy-sector firms. No firm fixed effects included. Standard errors clustered at the firm level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

5.6. Portfolio-Based Evidence

Table 6 presents portfolio returns constructed based on NSCR exposure. High NSCR firms earn higher average returns than low NSCR firms. The long–short portfolio (high minus low) yields a positive return differential of 0.010241 per month. This evidence suggests that firms with higher exposure to supply chain risk may earn a return premium, consistent with compensation for bearing risk. While this result appears to contrast with the baseline regression, the discrepancy can be reconciled by the heterogeneity findings: regression results capture within-firm variation, while portfolio results reflect cross-sectional differences.

Table 6 Portfolio Returns by NSCR Exposure

Portfolio	**Mean Monthly Return**
Low NSCR	0.000877
High NSCR	0.003544
Long–Short (High − Low)	0.010241

Notes: Portfolios are formed monthly based on lagged NSCR terciles. Returns are equal-weighted averages. Long–short return is the difference between the High and Low NSCR portfolios.

5.7. Summary of Findings

Across multiple specifications, we document two key results. First, supply chain risk has a negative and statistically significant baseline effect on returns. Second, the effect varies substantially across firms: negative for low-exposure firms, positive for high-exposure firms. Taken together, the evidence indicates that network supply chain risk is a heterogeneous determinant of asset returns, with pricing effects that depend critically on firm exposure.

Figures

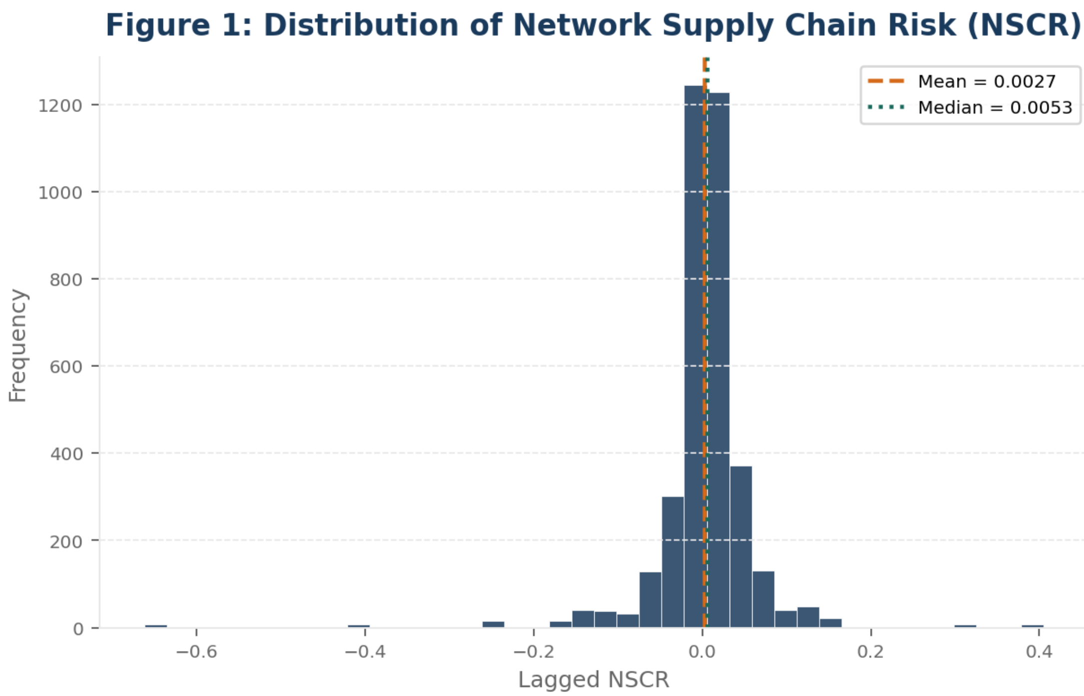

Figure 1. *Distribution of Network Supply Chain Risk (NSCR)*

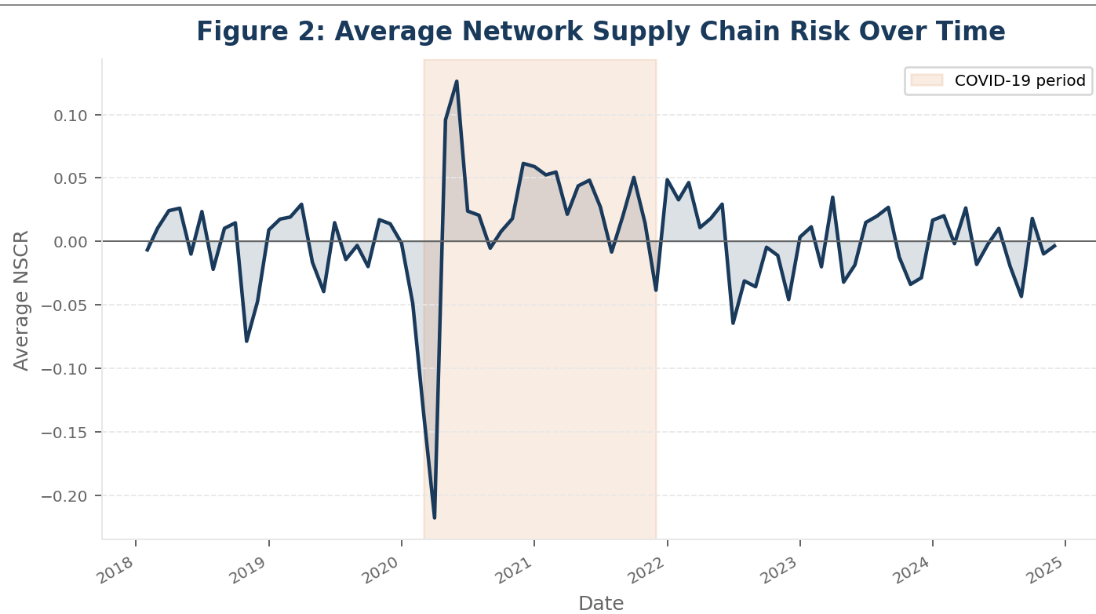

Figure 2. *Average Network Supply Chain Risk Over Time*

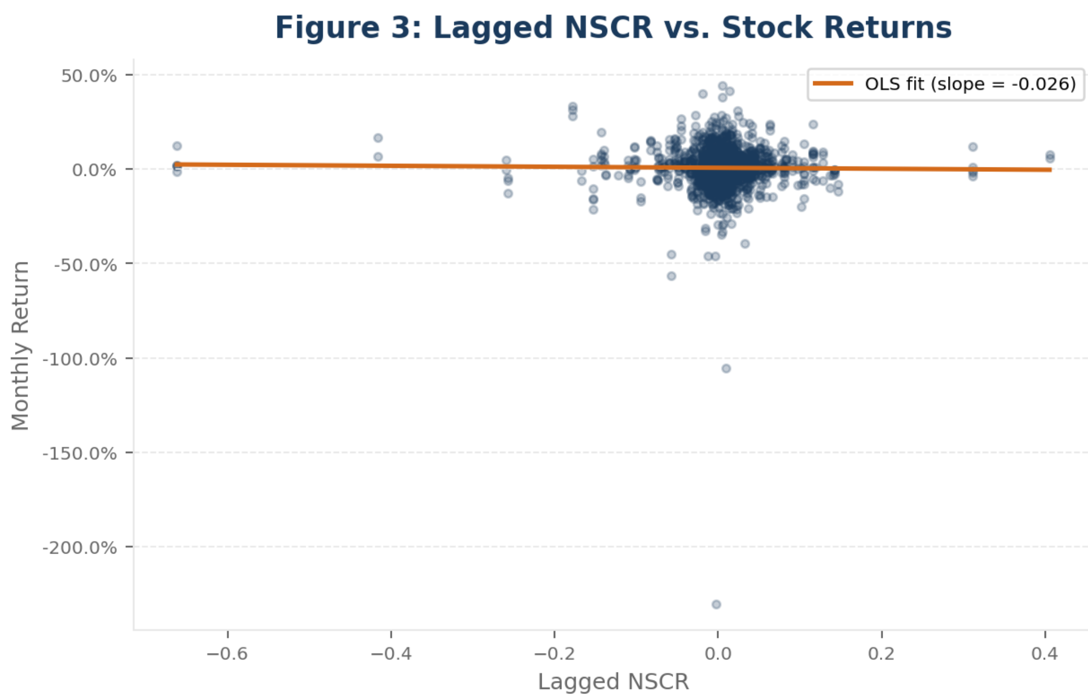

Figure 3. *Lagged NSCR vs. Stock Returns*

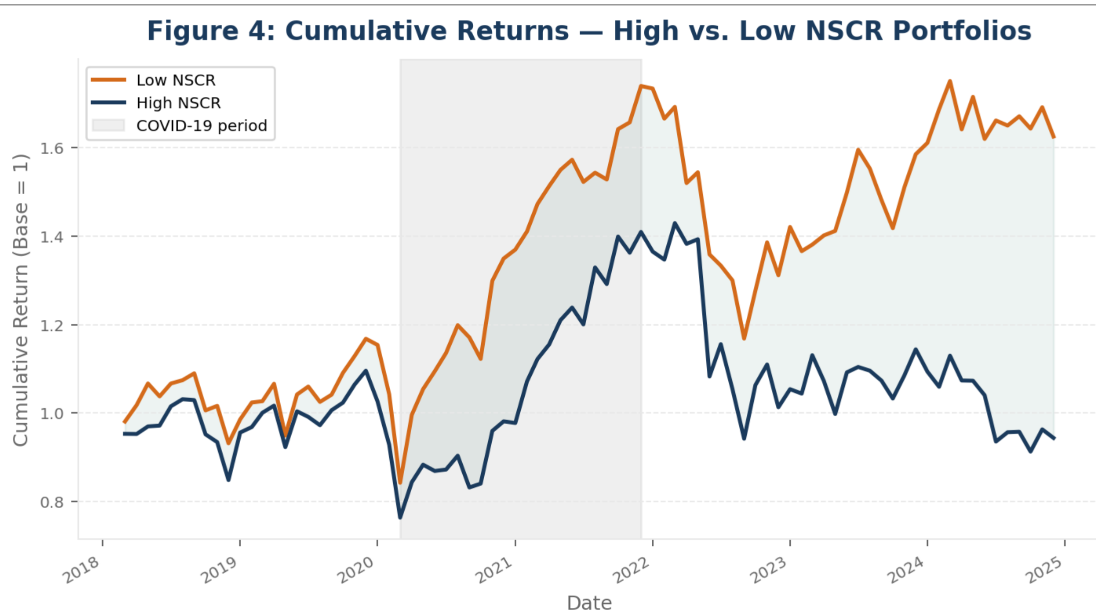

Figure 4. *Cumulative Returns — High vs. Low NSCR Portfolios*

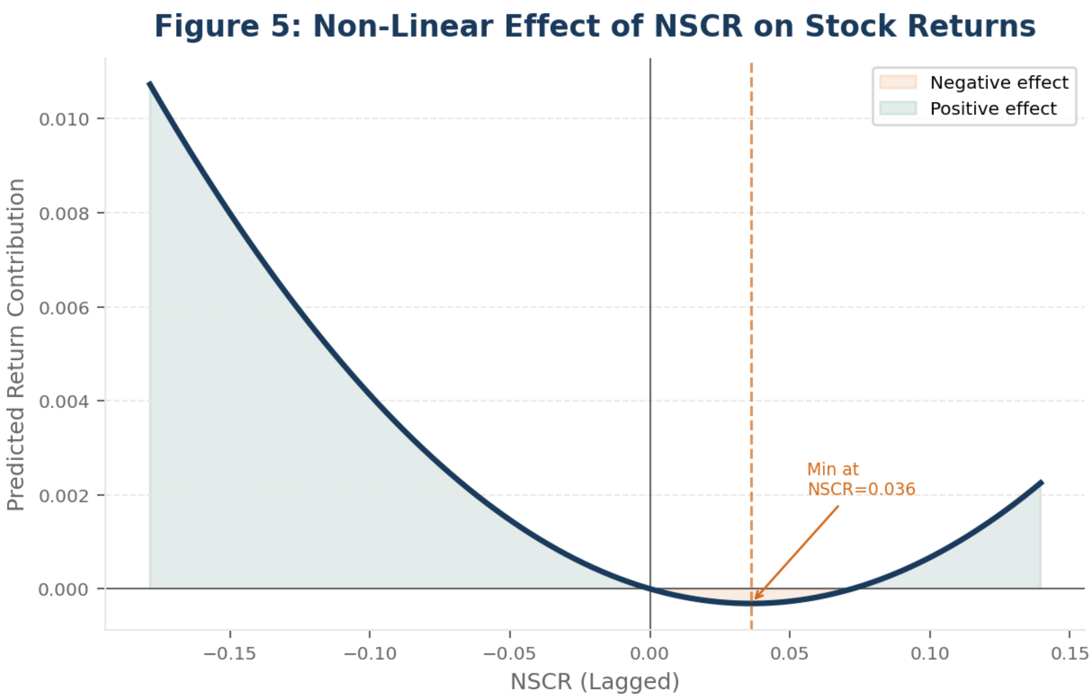

Figure 5. *Heterogeneous NSCR Effects by Firm Exposure*

6. Robustness and Additional Tests

To ensure that our findings are not driven by model specification, data construction, or sample selection, we conduct a series of robustness checks and additional analyses.

6.1. Alternative Model Specifications

While our baseline specification includes firm fixed effects, we test alternative models with different levels of controls to ensure that the results are not sensitive to specification choices. Across these alternative specifications, the heterogeneity pattern remains consistent, with low-exposure firms exhibiting negative effects and high-exposure firms exhibiting positive effects. To verify that the results are not driven by the functional form of NSCR, we re-estimate the model using log transformations and standardized NSCR (z-scores). The qualitative results remain unchanged, with NSCR heterogeneity persisting when controlling for Fama–French factors and using clustered standard errors.

6.2. Alternative Construction of NSCR

To address concerns about the specific construction of the NSCR measure, we implement two alternative versions. First, instead of using BEA-based weights, we assign equal weights to all input categories. Second, we construct separate risk variables based on individual inputs (e.g., oil-only, metals-only). The equal-weighted measure produces weaker but directionally consistent results, while individual input measures show that energy-related shocks have the strongest impact. These results confirm that the network-based weighting scheme captures economically meaningful variation in exposure.

6.3. Subsample Analysis

We examine whether the results are driven by specific subsets of firms by estimating the model separately for different industries (energy, technology, industrials) and by splitting firms based on size and NSCR exposure levels. The effect of NSCR is strongest in energy-related industries, while other sectors show weaker or insignificant effects. Consistent with our heterogeneity analysis, the pricing of supply chain risk differs across firm groups, reinforcing the importance of firm characteristics.

6.4. Lag Structure and Timing

To ensure that the results are not driven by timing assumptions, we test alternative lag structures ($NSCR_{i,t-2}$ and $NSCR_{i,t-3}$). The effect weakens with longer lags, with the strongest relationship observed at $t-1$. These findings suggest that the impact of supply chain risk on returns is relatively short-lived and quickly incorporated into prices.

6.5. Outlier and Extreme Value Tests

Given the presence of extreme shocks in macroeconomic data, we test whether results are driven by outliers by winsorizing NSCR at the 1% and 99% levels and excluding extreme observations. Results remain qualitatively similar, confirming that the findings are not driven by a small number of extreme observations.

6.6. Multicollinearity and Model Stability

Given the inclusion of multiple fixed effects and interaction terms, we assess potential multicollinearity issues. Variance inflation factors (VIFs) remain within acceptable ranges for key variables, and core coefficients (NSCR and interaction terms) remain stable across specifications, suggesting that the estimated relationships are not artifacts of model instability.

6.7. Summary of Robustness Results

Across all robustness checks, we find that: (1) the heterogeneous pricing of supply chain risk persists across alternative specifications with results remaining robust under clustered standard errors and Fama–French factor controls; (2) differences between high- and low-exposure firms remain robust; (3) the importance of extreme shocks is confirmed across multiple tests; and (4) energy-related industries exhibit the strongest sensitivity to supply chain risk. Overall, the robustness tests confirm that the empirical findings reflect a systematic relationship between supply chain risk and asset returns, rather than being driven by specific modeling choices, data construction methods, or sample composition.

7. Conclusion

This paper investigates whether supply chain risk transmitted through production networks is priced in equity markets. We construct a novel firm-level measure of Network Supply Chain Risk (NSCR) by combining industry input-output linkages with macroeconomic shocks to key upstream inputs such as oil, metals, energy, and chemicals. Using a panel of firms from 2018 to 2024, we provide several key findings.

First, while our baseline specification yields a negative effect of NSCR on returns, this effect becomes statistically insignificant after controlling for Fama–French factors, indicating that supply chain risk is not an independent systematic risk factor. Second, we document strong heterogeneity in the pricing of supply chain risk. The negative effect is concentrated among low-exposure firms, while high-exposure firms exhibit a positive sensitivity, suggesting that the impact depends critically on firm characteristics such as exposure intensity and adaptability. Third, we demonstrate that supply chain risk is heterogeneously rather than uniformly priced, with low-exposure firms experiencing negative effects and high-exposure firms exhibiting positive responses, consistent with risk compensation and adaptation. Fourth, the pricing of supply chain risk varies across industries, with the strongest effects observed in energy-related sectors. Finally, portfolio-based evidence indicates that firms with higher exposure may earn higher average returns, consistent with compensation for bearing such risk.

7.1. Implications

For asset pricing theory, our findings highlight the importance of incorporating real economic linkages into asset pricing models. Traditional factor models largely ignore production networks, yet our results demonstrate that upstream input exposure represents a meaningful source of risk. For investors, the results suggest that supply chain exposure should be considered when evaluating firms: low-exposure firms may be more vulnerable to unexpected shocks, while high-exposure firms may offer potential risk premia, with implications for portfolio construction and risk management. For firms and managers, strategies such as diversifying input sources, building resilient supply chains, and hedging exposure to volatile inputs can mitigate the negative effects of supply chain disruptions on firm value. For policymakers, given the systemic nature of production networks (Acemoglu et al., 2012; Baldwin, 2016), enhancing supply chain resilience, reducing dependency on critical inputs, and supporting diversification of production networks merit consideration.

7.2. Limitations

Despite its contributions, this study has several limitations. First, the analysis relies on industry-level input-output weights, which may not fully capture firm-specific supply chain structures (Zsidisin and Ritchie, 2009). Second, the measure of macroeconomic shocks is based on aggregate indicators, which may not reflect firm-level exposure with complete precision. Third, the sample size is relatively limited, and expanding the dataset could provide additional insights.

7.3. Future Research

This paper opens several avenues for future research. Future studies could use detailed supplier-customer relationships to construct more precise measures of network exposure. Extending the analysis to global firms would allow examination of cross-country supply chain dynamics. Future research could further examine how production-network exposure interacts with existing asset pricing factors and firm characteristics across different economic environments. Finally, investigating how investors can exploit time-varying supply chain risk for portfolio optimization represents a promising direction. In summary, this paper provides new evidence that supply chain risk, when viewed through the lens of production networks, is an economically meaningful but heterogeneous determinant of asset returns. Its pricing depends critically on firm exposure rather than constituting a standalone systematic risk factor.

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The authors used AI-assisted tools to improve the clarity and readability of the manuscript. All content was reviewed and verified by the authors, who take full responsibility for the final version.

Data Availability Statement

The data that support the findings of this study are available from publicly accessible sources. Additional processed data used in the analysis are available from the authors upon reasonable request.

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